

40-Bragg: Measuring Work Function of Non-Metallic Materials Using KPFM

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Abstract:

The work functions of twelve pre-prepared polymer films were measured via KPFM to map the triboelectric series.

Additional polymer films were fabricated to understand the effects of film thickness and substrate material on the measured work function

KPFM:

KPFM maps surface potential and measures contact potential difference, enabling the calculation of the work function of materials

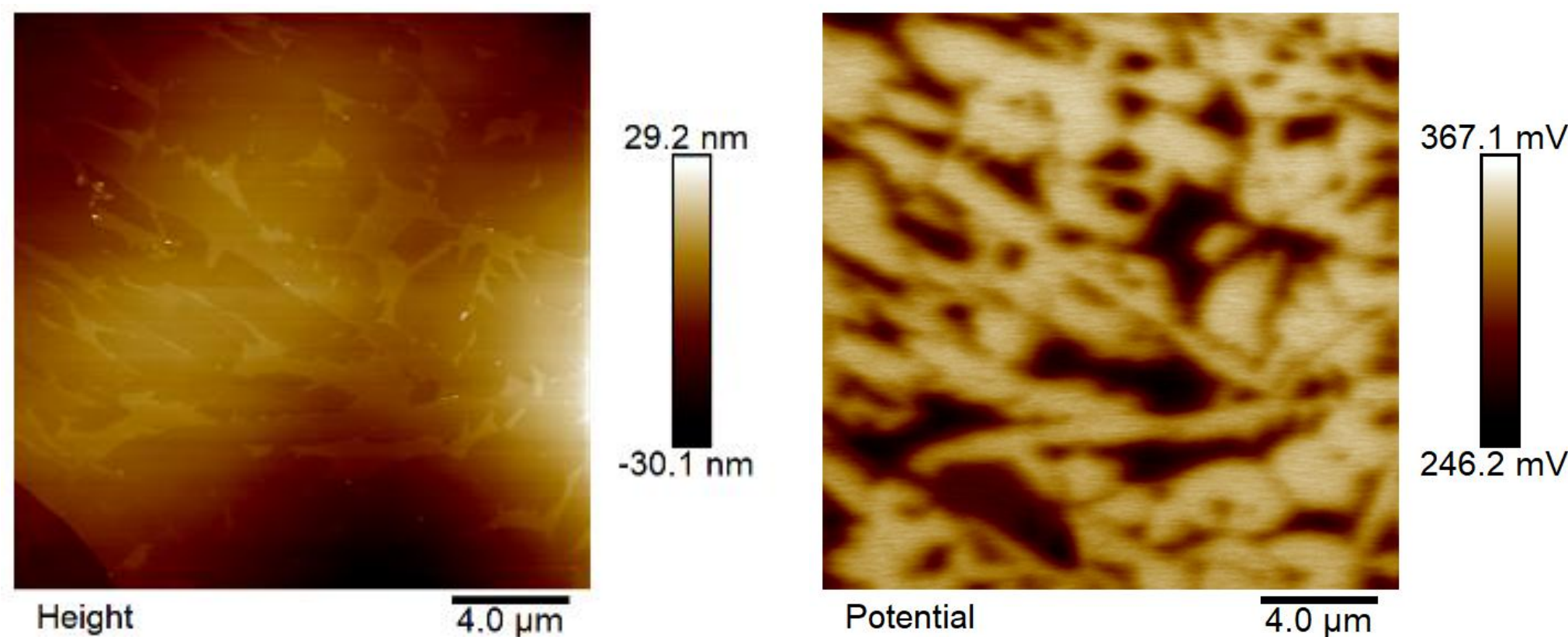


Fig.1 - Topography and surface potential of HOPG

Between layers: $\Delta_{CPD} = 80mV$, $\Delta_h = 2.6nm$

Effect of Substrate:

Work functions were measured for polished steel and gold-coated substrates **before and after deposition** of PVC films

$$\Delta\phi_{Substrates} = 0.2 \pm 0.1eV$$

$$\Delta\phi_{PVC} = 0.2 \pm 0.1eV$$

The difference in work functions was preserved after PVC deposition suggesting a **substrate dependant effect**

Future Work:

- Test the **reproducibility** of results
- Develop methods to **control film thickness**
- Evaluate the effects of **surface coatings**
- Investigate polymer **charging behaviour**

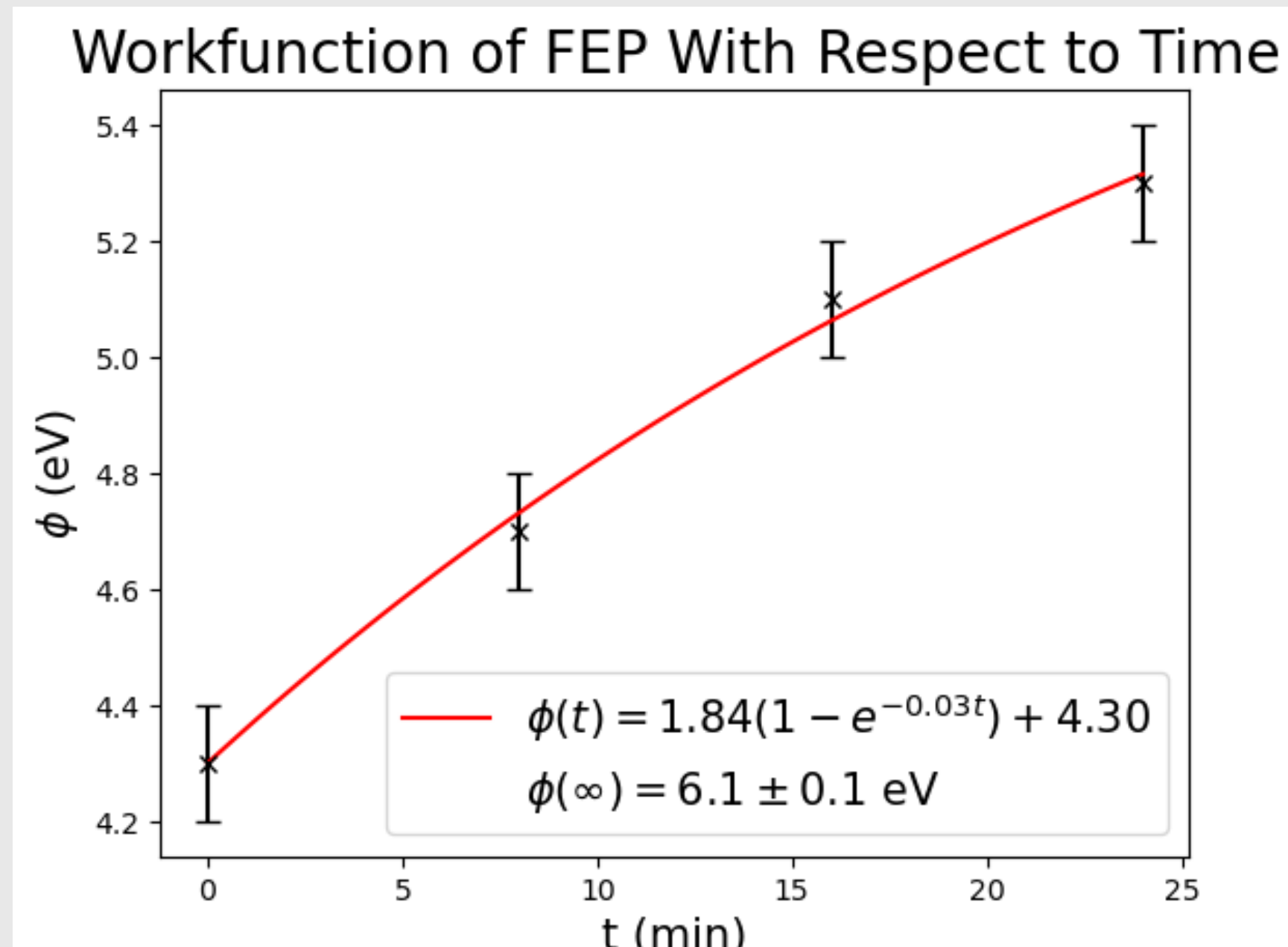


Fig.4 - Charging Behaviour of FEP

Sample Preparation:

Spin Coating:

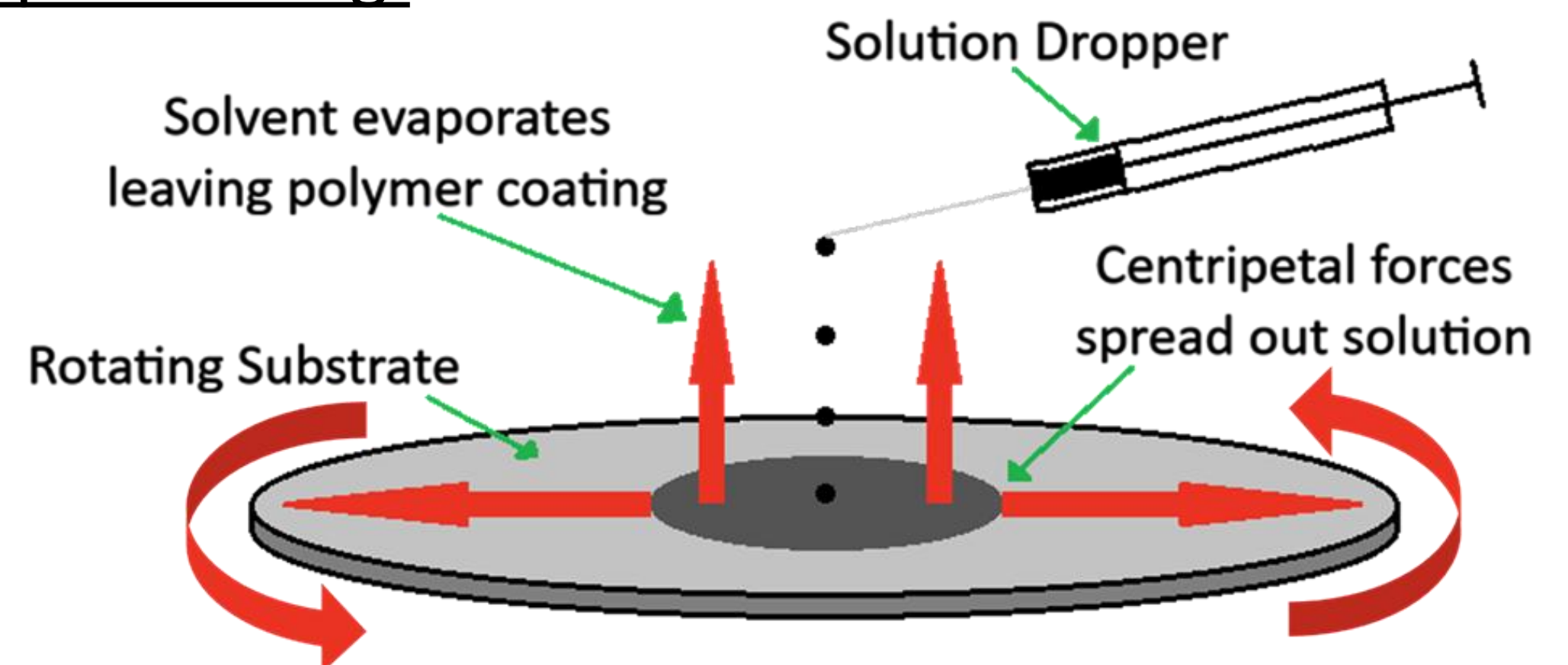


Fig.2 - Spin Coating apparatus

- Polymer **dissolved in solvent** and **spin-coated** onto polished metal substrate
- **Up to five layers deposited** to increase film thickness
- Each layer left to **air dry for five min**
- Multiple layers created visually rougher films
- Thickness of films could not be reliably obtained

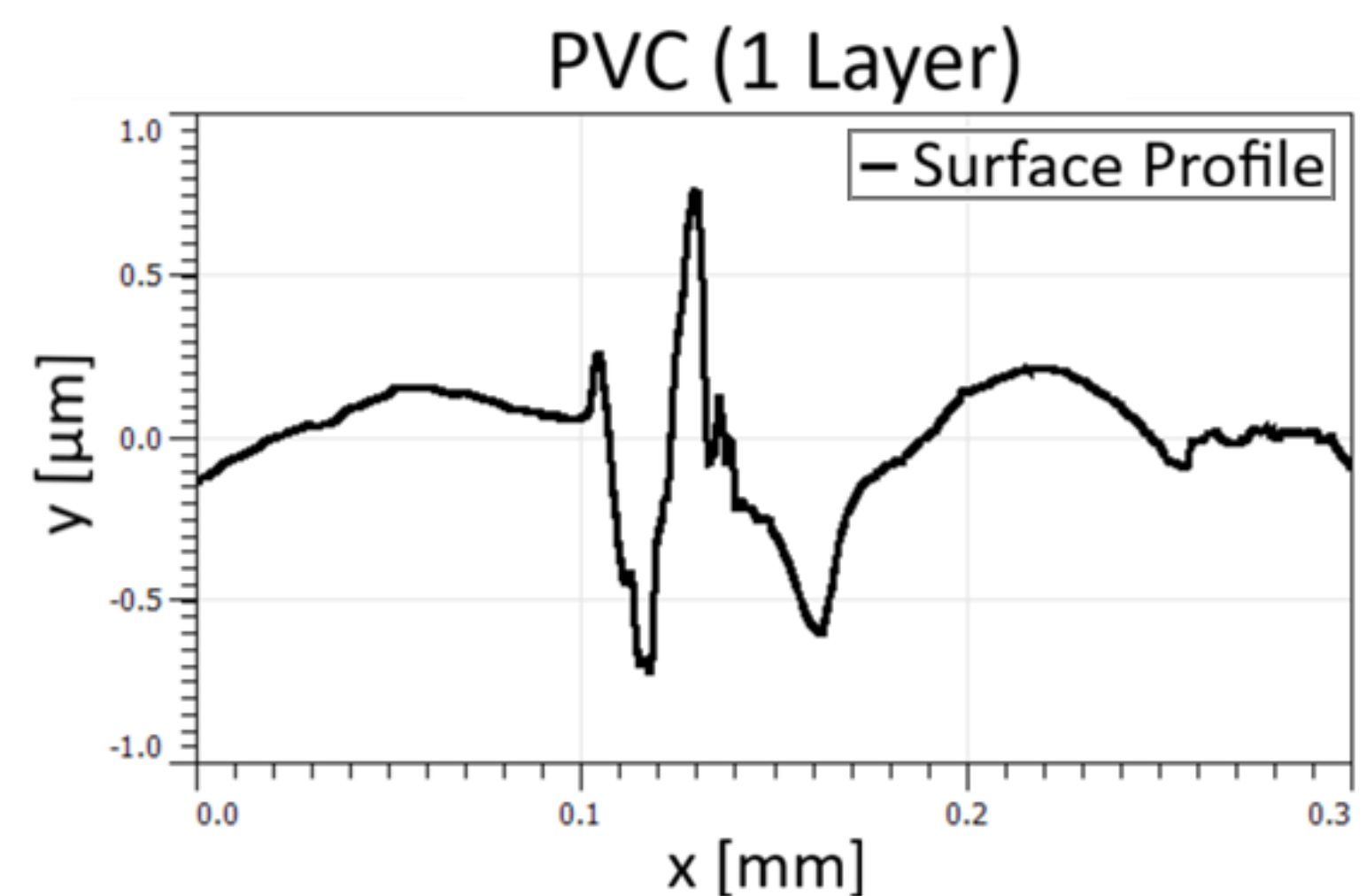


Fig.3 – Poor thickness measurement of PVC film

Heat Bonding:

- 17μm nylon sheets heat bonded onto substrate at **200°C for 10 min**
- Multiple layers used to increase thickness
- Poor adhesion prevented successful film fabrication

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